

Assingment—7

Task Description #1 (Syntax Errors – Missing Parentheses in Print Statement)

Bug code:

```
name="prathap"
```

```
print(name
```

output:

```
PS C:\Users\prath\vscode\github> python -u "c:\Users\prath\vscode\github\tempCodeRunnerFile.py"
File "c:\Users\prath\vscode\github\tempCodeRunnerFile.py", line 2
    print(name
    ^
SyntaxError: '(' was never closed
PS C:\Users\prath\vscode\github> 
```

prompt: find the error and fix the error and give correct code.

#The error in the code is that the print statement is not properly closed with a parenthesis. Here is the corrected code:

Correct code:

```
name = "prathap"
```

```
print(name)
```

```
n=10
```

```
    print("n is equal to 10")
```

```
else:
```

```
    print("n is not equal to 10")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
prathap
PS C:\Users\prath\.vscode\github> █
```

Task Description #2 (Incorrect condition in an If Statement)

Bug code:

```
n=10
if n=10:
    print("n is equal to 10")
else:
    print("n is not equal to 10")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    if n=10:
        ^^^^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

Prompt: find the error and fix the error and give correct code.

Correct code:

```
n = 10
if n == 10:
    print("n is equal to 10")
else:
    print("n is not equal to 10")
```

Output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
n is equal to 10
PS C:\Users\prath\.vscode\github> █
```

Task Description #3 (Runtime Error – File Not Found)

Bug code:

```
def read_file(filename):
    with open(filename, &#39;r&#39;) as f:
        return f.read()
print(read_file(&quot;nonexistent.txt&quot;))
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    with open(filename, &#39;r&#39;) as f:
    ^^^^^
IndentationError: expected an indented block after function definition on line 1
```

prompt: solve the error and give the correct code and explain that

correct code:

#The error in the provided code occurs because the file "nonexistent.txt" does not exist, which raises a FileNotFoundError. To handle this error gracefully, we can use a try-except block to catch the exception and provide a meaningful message instead of crashing the program.

#Here is the corrected code with error handling:

```
def read_file(filename):
    try:
```

```
with open(filename, 'r') as f:

    return f.read()

except FileNotFoundError:

    return f"Error: The file '{filename}' does not exist."

print(read_file("da.txt"))
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
how are you.
PS C:\Users\prath\.vscode\github> |
```

Task Description #4 (Calling a Non-Existent Method)

Task: Give a class where a non-existent method is called (e.g., `obj.undefined_method()`). Use AI to debug and fix.

Bug code:

```
class Car:

def start(self):

return "Car started"

my_car = Car()

print(my_car.drive())
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 2
    def start(self):
    ^^^
IndentationError: expected an indented block after class definition on line 1
```

prompt: find the error and fix the error and give correct code.

#The error in the code is that the method 'start' is not properly indented and the method 'drive' is called instead of 'start'. Here is the corrected code:

Correct code:

```
class Car:
```

```
def start(self):  
    return "Car started"
```

```
my_car = Car()  
print(my_car.start())
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
Car started  
PS C:\Users\prath\.vscode\github> █
```

Task Description #5 (TypeError – Mixing Strings and Integers in Addition)

Task: Provide code that adds an integer and string ('5' + 2) causing a TypeError.

Bug code:

```
x="5"  
  
y=2  
  
print(x+y)
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
Traceback (most recent call last):  
  File "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py", line 3, in <module>  
    print(x+y)  
          ~~~  
TypeError: can only concatenate str (not "int") to str
```

prompt: find the error and fix the error and give correct code.

#The error in the code is that 'x' is a string and 'y' is an integer, which causes a TypeError when trying to add them. To fix this, we can convert 'x' to an integer before adding. Here is the corrected code:

Correct code:

```
x = "5"  
  
y = 2  
  
print(int(x) + y)
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
7  
PS C:\Users\prath\.vscode\github> █
```